

Hotel Reviews Data Analysis

AUTHOR

Team Details Below

Team Information

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1. Introduction

This project analyzes hotel review datasets to extract insights about customer satisfaction, ratings, and patterns.

loading the library

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.0      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.2      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.1
— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

getting and binding the datasets

```
df1 <- read_csv("7282_1.csv")
```

Rows: 35912 Columns: 19

— Column specification —

Delimiter: ","

chr (12): address, categories, city, country, name, postalCode, province, r...

dbl (3): latitude, longitude, reviews.rating

```
lgl (2): reviews.doRecommend, reviews.id
dtm (2): reviews.date, reviews.dateAdded
```

• Use `spec()` to retrieve the full column specification for this data.
• Specify the column types or set `show_col_types = FALSE` to quiet this message.

```
df2 <- read_csv("Datafiniti_Hotel_Reviews.csv")
```

Rows: 10000 Columns: 25

— Column specification —

Delimiter: ","

chr (19): id, address, categories, primaryCategories, city, country, keys, ...

dbl (3): latitude, longitude, reviews.rating

dtm (3): dateAdded, dateUpdated, reviews.date

• Use `spec()` to retrieve the full column specification for this data.
• Specify the column types or set `show_col_types = FALSE` to quiet this message.

```
df3 <- read_csv("Datafiniti_Hotel_Reviews_Jun19.csv")
```

Rows: 10000 Columns: 26

— Column specification —

Delimiter: ","

chr (19): id, address, categories, primaryCategories, city, country, keys, ...

dbl (3): latitude, longitude, reviews.rating

lgl (1): reviews.dateAdded

dtm (3): dateAdded, dateUpdated, reviews.date

• Use `spec()` to retrieve the full column specification for this data.
• Specify the column types or set `show_col_types = FALSE` to quiet this message.

```
df <- bind_rows(df1, df2, df3)
```

```
print(head(df))
```

A tibble: 6 × 28

	address	categories	city	country	latitude	longitude	name	postalCode	province
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
2	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
3	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
4	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
5	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
6	Riviera...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA

i 19 more variables: reviews.date <dtm>, reviews.dateAdded <dtm>,
reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>, keys <chr>,
reviews.dateSeen <chr>, reviews.sourceURLs <chr>, sourceURLs <chr>,
websites <chr>

Setup Column

```
num_col <- df %>% select(where(is.numeric)) %>% names() %>% .[1]
cat_col <- df %>% select(where(is.character)) %>% names() %>% .[1]

print(num_col)
```

```
[1] "latitude"
```

```
print(cat_col)
```

```
[1] "address"
```

```
# Gets column names of the dataset (information about data structure)
print(names(df))
```

```
[1] "address"           "categories"         "city"
[4] "country"           "latitude"           "longitude"
[7] "name"              "postalCode"         "province"
[10] "reviews.date"       "reviews.dateAdded"  "reviews.doRecommend"
[13] "reviews.id"         "reviews.rating"      "reviews.text"
[16] "reviews.title"      "reviews.userCity"    "reviews.username"
[19] "reviews.userProvince" "id"                  "dateAdded"
[22] "dateUpdated"        "primaryCategories"   "keys"
[25] "reviews.dateSeen"   "reviews.sourceURLs"  "sourceURLs"
[28] "websites"
```

```
# Shows structure of the dataset
print(str(df))
```

```
spc_tbl_ [55,912 × 28] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ address          : chr [1:55912] "Riviera San Nicol 11/a" "Riviera San Nicol 11/a"
"Riviera San Nicol 11/a" "Riviera San Nicol 11/a" ...
 $ categories       : chr [1:55912] "Hotels" "Hotels" "Hotels" "Hotels" ...
 $ city             : chr [1:55912] "Mableton" "Mableton" "Mableton" "Mableton" ...
 $ country          : chr [1:55912] "US" "US" "US" "US" ...
 $ latitude         : num [1:55912] 45.4 45.4 45.4 45.4 45.4 ...
 $ longitude        : num [1:55912] 12.4 12.4 12.4 12.4 12.4 ...
 $ name             : chr [1:55912] "Hotel Russo Palace" "Hotel Russo Palace" "Hotel Russo
Palace" "Hotel Russo Palace" ...
 $ postalCode       : chr [1:55912] "30126" "30126" "30126" "30126" ...
 $ province         : chr [1:55912] "GA" "GA" "GA" "GA" ...
 $ reviews.date     : POSIXct[1:55912], format: "2013-09-22 00:00:00" "2015-04-03 00:00:00"
...
 $ reviews.dateAdded : POSIXct[1:55912], format: "2016-10-24 00:00:25" "2016-10-24 00:00:25"
...
 $ reviews.doRecommend : logi [1:55912] NA NA NA NA NA NA ...
 $ reviews.id         : logi [1:55912] NA NA NA NA NA NA ...
 $ reviews.rating     : num [1:55912] 4 5 5 5 5 5 4 4 3 4 ...
 $ reviews.text       : chr [1:55912] "Pleasant 10 min walk along the sea front to the Water
Bus. restaurants etc. Hotel was comfortable breakfast was" | __truncated__ "Really lovely
hotel. Stayed on the very top floor and were surprised by a Jacuzzi bath we didn't know we
```

```
were g"| __truncated__ "Ett mycket bra hotell. Det som drog ner betyget var att vi fick ett
rum under taksarna dr det endast var full s"| __truncated__ "We stayed here for four nights in
October. The hotel staff were welcoming, friendly and helpful. Assisted in bo"| __truncated__
...
$ reviews.title      : chr [1:55912] "Good location away from the crouds" "Great hotel with
Jacuzzi bath!" "Lugnt l  ge" "Good location on the Lido." ...
$ reviews.userCity    : chr [1:55912] NA NA NA NA ...
$ reviews.username    : chr [1:55912] "Russ (kent)" "A Traveler" "Maud" "Julie" ...
$ reviews.userProvince: chr [1:55912] NA NA NA NA ...
$ id                  : chr [1:55912] NA NA NA NA ...
$ dateAdded           : POSIXct[1:55912], format: NA NA ...
$ dateUpdated         : POSIXct[1:55912], format: NA NA ...
$ primaryCategories   : chr [1:55912] NA NA NA NA ...
$ keys                : chr [1:55912] NA NA NA NA ...
$ reviews.dateSeen    : chr [1:55912] NA NA NA NA ...
$ reviews.sourceURLs  : chr [1:55912] NA NA NA NA ...
$ sourceURLs          : chr [1:55912] NA NA NA NA ...
$ websites            : chr [1:55912] NA NA NA NA ...
- attr(*, "spec")=
.. cols(
..   address = col_character(),
..   categories = col_character(),
..   city = col_character(),
..   country = col_character(),
..   latitude = col_double(),
..   longitude = col_double(),
..   name = col_character(),
..   postalCode = col_character(),
..   province = col_character(),
..   reviews.date = col_datetime(format = ""),
..   reviews.dateAdded = col_datetime(format = ""),
..   reviews.doRecommend = col_logical(),
..   reviews.id = col_logical(),
..   reviews.rating = col_double(),
..   reviews.text = col_character(),
..   reviews.title = col_character(),
..   reviews.userCity = col_character(),
..   reviews.username = col_character(),
..   reviews.userProvince = col_character()
.. )
- attr(*, "problems")=<externalptr>
NULL
```

```
# Gives summary statistics of the dataset
print(summary(df))
```

address	categories	city	country
Length:55912	Length:55912	Length:55912	Length:55912
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

latitude	longitude	name	postalCode
Min. : -25.44	Min. : -166.56	Length:55912	Length:55912
1st Qu.: 33.51	1st Qu.: -110.93	Class :character	Class :character
Median : 37.42	Median : -87.95	Mode :character	Mode :character
Mean : 36.83	Mean : -89.82		
3rd Qu.: 41.45	3rd Qu.: -78.98		
Max. : 70.13	Max. : 115.16		
NA's :86	NA's :86		
province	reviews.date	reviews.dateAdded	
Length:55912	Min. :2002-05-16 00:00:00	Min. :2015-01-28 14:40:46	
Class :character	1st Qu.:2015-03-04 00:00:00	1st Qu.:2016-10-27 22:10:18	
Mode :character	Median :2015-11-29 00:00:00	Median :2016-11-05 20:23:11	
	Mean :2015-06-20 03:04:25	Mean :2016-10-30 09:03:22	
	3rd Qu.:2016-06-14 00:00:00	3rd Qu.:2016-11-17 15:07:50	
	Max. :2019-01-30 00:00:00	Max. :2017-04-20 01:37:39	
	NA's :259	NA's :20000	
reviews.doRecommend	reviews.id	reviews.rating	reviews.text
Mode:logical	Mode:logical	Min. : 0.000	Length:55912
NA's:55912	NA's:55912	1st Qu.: 3.000	Class :character
		Median : 4.000	Mode :character
		Mean : 3.879	
		3rd Qu.: 5.000	
		Max. :10.000	
		NA's :862	
reviews.title	reviews.userCity	reviews.username	reviews.userProvince
Length:55912	Length:55912	Length:55912	Length:55912
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

id	dateAdded	dateUpdated	
Length:55912	Min. :2014-10-24 13:52:43	Min. :2018-01-01 00:00:43	
Class :character	1st Qu.:2016-05-20 01:56:27	1st Qu.:2018-03-23 17:12:04	
Mode :character	Median :2017-03-04 12:22:32	Median :2018-10-21 21:15:25	
	Mean :2017-02-15 22:37:13	Mean :2018-10-08 02:38:42	
	3rd Qu.:2017-12-21 00:00:53	3rd Qu.:2019-04-02 05:26:23	
	Max. :2018-12-28 06:33:31	Max. :2019-05-20 23:55:47	
	NA's :35912	NA's :35912	
primaryCategories	keys	reviews.dateSeen	reviews.sourceURLs
Length:55912	Length:55912	Length:55912	Length:55912
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

sourceURLs	websites
Length:55912	Length:55912
Class :character	Class :character
Mode :character	Mode :character

```
# Adds a new column with constant value
df$new_col <- 1
print(df)
```

A tibble: 55,912 × 29

	address	categories	city	country	latitude	longitude	name	postalCode	province
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
2	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
3	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
4	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
5	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
6	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
7	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
8	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
9	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
10	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA

i 55,902 more rows

```
# i 20 more variables: reviews.date <dtm>, reviews.dateAdded <dtm>,
# reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
# reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
# reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
# dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>, keys <chr>,
# reviews.dateSeen <chr>, reviews.sourceURLs <chr>, sourceURLs <chr>, ...
```

```
# Adds a new column using first numeric column (safe)
num_col <- names(df)[sapply(df, is.numeric)][1]
df$double_val <- df[[num_col]] * 2
print(df)
```

A tibble: 55,912 × 30

	address	categories	city	country	latitude	longitude	name	postalCode	province
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
2	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
3	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
4	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
5	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
6	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
7	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
8	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
9	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
10	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA

i 55,902 more rows

```
# i 21 more variables: reviews.date <dtm>, reviews.dateAdded <dtm>,
# reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
# reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
# reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
```

```
# dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>, keys <chr>,
# reviews.dateSeen <chr>, reviews.sourceURLs <chr>, sourceURLs <chr>, ...
```

```
# Deletes the added column
df$new_col <- NULL
print(df)
```

```
# A tibble: 55,912 × 29
```

	address	categories	city	country	latitude	longitude	name	postalCode	province
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
2	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
3	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
4	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
5	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
6	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
7	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
8	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
9	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA
10	Rivier...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126	GA

```
# i 55,902 more rows
```

```
# i 20 more variables: reviews.date <dtm>, reviews.dateAdded <dtm>,
# reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
# reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
# reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
# dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>, keys <chr>,
# reviews.dateSeen <chr>, reviews.sourceURLs <chr>, sourceURLs <chr>, ...
```

```
# Renames first column safely
colnames(df)[1] <- "first_column"
print(df)
```

```
# A tibble: 55,912 × 29
```

	first_column	categories	city	country	latitude	longitude	name	postalCode
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>	<chr>
1	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
2	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
3	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
4	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
5	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
6	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
7	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
8	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
9	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126
10	Riviera San Nic...	Hotels	Mabl...	US	45.4	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 21 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Reorders columns (moves numeric column to first position)
df <- df[, c(num_col, setdiff(names(df), num_col))]
print(df)
```

```
# A tibble: 55,912 × 29
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 21 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Gets subset of rows where numeric value is above mean
mean_val <- mean(df[[num_col]], na.rm = TRUE)
print(df[df[[num_col]] > mean_val, ])
```

```
# A tibble: 30,864 × 29
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 30,854 more rows
```

```
# i 21 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Converts numeric column into factor (safe, no missing column issue)
df$num_factor <- as.factor(df[[num_col]])
```



```
print(df)
```

```
# A tibble: 55,912 × 30
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 22 more variables: province <chr>, reviews.date <dtm>,  
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,  
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,  
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,  
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,  
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Recodes numeric values into categories (safe)  
df$category <- ifelse(df[[num_col]] > mean_val, "High", "Low")  
print(df)
```

```
# A tibble: 55,912 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,  
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,  
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,  
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,  
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,  
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

Selection & Filtering

```
# Selects only numeric columns  
print(select(df, where(is.numeric)))
```

```
# A tibble: 55,912 × 4
  latitude longitude reviews.rating double_val
    <dbl>      <dbl>          <dbl>      <dbl>
1    45.4      12.4            4        90.8
2    45.4      12.4            5        90.8
3    45.4      12.4            5        90.8
4    45.4      12.4            5        90.8
5    45.4      12.4            5        90.8
6    45.4      12.4            5        90.8
7    45.4      12.4            4        90.8
8    45.4      12.4            4        90.8
9    45.4      12.4            3        90.8
10   45.4      12.4            4        90.8
# i 55,902 more rows
```

```
# Selects only character (text) columns
print(select(df, where(is.character)))
```

```
# A tibble: 55,912 × 20
  first_column categories city country name postalCode province reviews.text
    <chr>          <chr>    <chr> <chr>  <chr> <chr>      <chr>    <chr>
1 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Pleasant 10...
2 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Really love...
3 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Ett mycket ...
4 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We stayed h...
5 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We stayed h...
6 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We loved st...
7 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Lovely view...
8 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    ottimo sugg...
9 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Gnstiger Au...
10 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Lidoen er p...
# i 55,902 more rows
# i 12 more variables: reviews.title <chr>, reviews.userCity <chr>,
# reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
# primaryCategories <chr>, keys <chr>, reviews.dateSeen <chr>,
# reviews.sourceURLs <chr>, sourceURLs <chr>, websites <chr>, category <chr>
```

```
# Filters rows where numeric column is not missing
print(filter(df, !is.na(.data[[num_col]])))
```

```
# A tibble: 55,826 × 31
  latitude first_column categories city country longitude name postalCode
    <dbl> <chr>          <chr>    <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
```

```
# i 55,816 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Sorts data in ascending order based on numeric column
print(arrange(df, .data[[num_col]]))
```

```
# A tibble: 55,912 × 31
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr>   <chr> <chr>      <dbl> <chr> <chr>
1    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
2    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
3    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
4    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
5    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
6    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
7    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
8    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
9    -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
10   -25.4 Rua Desembargad... Hotels  Curi... US        -49.3 Merc... <NA>
```

```
# i 55,902 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Sorts data in descending order based on numeric column
print(arrange(df, desc(.data[[num_col]])))
```

```
# A tibble: 55,912 × 31
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr>   <chr> <chr>      <dbl> <chr> <chr>
1     70.1 6011 Lake Rd    Hotels,Lo... Kakt... US        -144. Wald... 99747
2     64.8 575 1st Ave     Hotels an... Fair... US        -148. Spri... 99701
3     64.8 575 1st Ave     Hotels an... Fair... US        -148. Spri... 99701
4     64.8 575 1st Ave     Hotels an... Fair... US        -148. Spri... 99701
5     64.8 575 1st Ave     Hotels an... Fair... US        -148. Spri... 99701
6     64.8 4888 Old Airpor... Hotels,Ho... Fair... US        -148. Gold... 99709
7     63.9 18 Hematite Pl   Hotels,Ho... Healy US        -149. Aspe... 99743
8     63.9 18 Hematite Pl   Hotels,Ho... Healy US        -149. Aspe... 99743
9     63.9 18 Hematite Pl   Hotels,Ho... Healy US        -149. Aspe... 99743
10    63.9 18 Hematite Pl   Hotels,Ho... Healy US        -149. Aspe... 99743
```

```
# i 55,902 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
```

```
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Removes duplicate rows
print(distinct(df))
```

```
# A tibble: 55,698 × 31
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr>   <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
# i 55,688 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Selects first 10 rows
print(slice(df, 1:10))
```

```
# A tibble: 10 × 31
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr>   <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>,
# sourceURLs <chr>, websites <chr>, double_val <dbl>, num_factor <fct>, ...
```

```
# Randomly selects 5 rows
```

```
print(sample_n(df, 5))
```

```
# A tibble: 5 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	39.3	711 Eastern Aven...	Hotel,Hot...	Balt...	US	-76.6	Pier...	21202
2	30.4	850 Bayview Ave	Hotels,Ca...	Bilo...	US	-88.9	Ip C...	39530
3	29.5	4934 N W Loop 410	Hotels	San ...	US	-98.6	Fies...	78229
4	29.5	4934 N W Loop 410	Hotels	San ...	US	-98.6	Fies...	78229
5	48.4	2300 Market St	Hotels,Lo...	Moun...	US	-122.	Best...	98273

```
# i 23 more variables: province <chr>, reviews.date <dtm>,  
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,  
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,  
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,  
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,  
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>,  
# sourceURLs <chr>, websites <chr>, double_val <dbl>, num_factor <fct>, ...
```

```
# Filters first 10 rows using row number  
print(df %>% filter(row_number() <= 10))
```

```
# A tibble: 10 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 23 more variables: province <chr>, reviews.date <dtm>,  
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,  
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,  
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,  
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,  
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>,  
# sourceURLs <chr>, websites <chr>, double_val <dbl>, num_factor <fct>, ...
```

```
# Filters rows where numeric column is not NA (using pipe)  
print(df %>% filter(!is.na(.data[[num_col]])))
```

```
# A tibble: 55,826 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```

6      45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
7      45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
8      45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
9      45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
10     45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
# i 55,816 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...

```

Mutations

```

# Adds a new column with constant value 1
print(mutate(df, constant = 1))

```

```

# A tibble: 55,912 × 32
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr> <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
# i 55,902 more rows
# i 24 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...

```

```

# Adds a column with row numbers
print(mutate(df, row_id = row_number()))

```

```

# A tibble: 55,912 × 32
  latitude first_column categories city country longitude name postalCode
  <dbl> <chr>          <chr> <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels Mabl... US      12.4 Hote... 30126

```

```

9      45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
10     45.4 Riviera San Nic... Hotels      Mabl... US      12.4 Hote... 30126
# i 55,902 more rows
# i 24 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...

```

```

# Creates a column with values doubled
print(mutate(df, double_val = .data[[num_col]] * 2))

```

```

# A tibble: 55,912 × 31
  latitude first_column      categories city  country longitude name  postalCode
  <dbl> <chr>              <chr>   <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
# i 55,902 more rows
# i 23 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...

```

```

# Creates a column with square root of values
print(mutate(df, sqrt_val = sqrt(abs(.data[[num_col]]))))

```

```

# A tibble: 55,912 × 32
  latitude first_column      categories city  country longitude name  postalCode
  <dbl> <chr>              <chr>   <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels    Mabl... US      12.4 Hote... 30126
# i 55,902 more rows
# i 24 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,

```



```
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Creates a column with log values
print(mutate(df, log_val = log(abs(.data[[num_col]]) + 1)))
```

```
# A tibble: 55,912 × 32
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 24 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Creates a flag column based on condition
print(df %>% mutate(flag = ifelse(.data[[num_col]] > 0, "Positive", "Other")))
```

```
# A tibble: 55,912 × 32
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 24 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```



```
# Categorizes values using multiple conditions
print(df %>% mutate(category = case_when(.data[[num_col]] > 0 ~ "Positive", TRUE ~ "Other")))
```

```
# A tibble: 55,912 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Converts numeric column to character
print(df %>% mutate(char_col = as.character(.data[[num_col]])))
```

```
# A tibble: 55,912 × 32
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 24 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Converts column to numeric type
print(df %>% mutate(num_new = as.numeric(.data[[num_col]])))
```

```
# A tibble: 55,912 × 32
  latitude first_column categories city country longitude name postalCode
    <dbl> <chr>          <chr>    <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
# i 55,902 more rows
# i 24 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Converts column to factor (categorical)
print(df %>% mutate(factor_col = as.factor(.data[[num_col]])))
```

```
# A tibble: 55,912 × 32
  latitude first_column categories city country longitude name postalCode
    <dbl> <chr>          <chr>    <chr> <chr>      <dbl> <chr> <chr>
1    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
2    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
3    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
4    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
5    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
6    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
7    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
8    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
9    45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
10   45.4 Riviera San Nic... Hotels  Mabl... US        12.4 Hote... 30126
# i 55,902 more rows
# i 24 more variables: province <chr>, reviews.date <dtm>,
#   reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
#   reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
#   reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
#   id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
#   keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

Summary

```
# Counts total number of rows in the dataset
print(df %>% summarise(count = n()))
```

```
# A tibble: 1 × 1
  count
```

```
<int>
1 55912
```

```
# Calculates mean (average) of numeric column
print(df %>% summarise(mean_val = mean(.data[[num_col]], na.rm = TRUE)))
```

```
# A tibble: 1 × 1
  mean_val
  <dbl>
1    36.8
```

```
# Finds maximum value in numeric column
print(df %>% summarise(max_val = max(.data[[num_col]], na.rm = TRUE)))
```

```
# A tibble: 1 × 1
  max_val
  <dbl>
1    70.1
```

```
# Finds minimum value in numeric column
print(df %>% summarise(min_val = min(.data[[num_col]], na.rm = TRUE)))
```

```
# A tibble: 1 × 1
  min_val
  <dbl>
1   -25.4
```

```
# Calculates standard deviation (spread of data)
print(df %>% summarise(sd_val = sd(.data[[num_col]], na.rm = TRUE)))
```

```
# A tibble: 1 × 1
  sd_val
  <dbl>
1    6.70
```

```
# Counts frequency of each value in numeric column
print(df %>% count(.data[[num_col]]))
```

```
# A tibble: 4,125 × 2
  latitude      n
  <dbl> <int>
1   -25.4     55
2   -8.70     56
3   -6.20     83
4    9.49     24
5   13.4      11
6   15.5      80
7   16.8      57
8   18.5      51
9   19.4       1
```

```
10    19.4    73
# i 4,115 more rows
```

```
# Removes rows with missing values
print(df %>% drop_na())
```

```
# A tibble: 0 × 31
# i 31 variables: latitude <dbl>, first_column <chr>, categories <chr>,
#   city <chr>, country <chr>, longitude <dbl>, name <chr>, postalCode <chr>,
#   province <chr>, reviews.date <dtm>, reviews.dateAdded <dtm>,
#   reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
#   reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
#   reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
#   dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>, ...
```

```
# Selects only numeric columns
print(df %>% select(where(is.numeric)))
```

```
# A tibble: 55,912 × 4
  latitude longitude reviews.rating double_val
    <dbl>      <dbl>          <dbl>      <dbl>
1    45.4      12.4            4        90.8
2    45.4      12.4            5        90.8
3    45.4      12.4            5        90.8
4    45.4      12.4            5        90.8
5    45.4      12.4            5        90.8
6    45.4      12.4            5        90.8
7    45.4      12.4            4        90.8
8    45.4      12.4            4        90.8
9    45.4      12.4            3        90.8
10   45.4      12.4            4        90.8
# i 55,902 more rows
```

```
# Selects only character (text) columns
print(df %>% select(where(is.character)))
```

```
# A tibble: 55,912 × 20
  first_column categories city country name postalCode province reviews.text
  <chr>          <chr>    <chr> <chr>  <chr> <chr>      <chr>    <chr>
1 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Pleasant 10...
2 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Really love...
3 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Ett mycket ...
4 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We stayed h...
5 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We stayed h...
6 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    We loved st...
7 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Lovely view...
8 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    ottimo sogg...
9 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Gnstiger Au...
10 Riviera San ... Hotels  Mabl... US    Hote... 30126    GA    Lidoen er p...
# i 55,902 more rows
# i 12 more variables: reviews.title <chr>, reviews.userCity <chr>,
#   reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
```

```
# primaryCategories <chr>, keys <chr>, reviews.dateSeen <chr>,
# reviews.sourceURLs <chr>, sourceURLs <chr>, websites <chr>, category <chr>
```

```
# Removes duplicate rows from dataset
print(df %>% distinct())
```

```
# A tibble: 55,698 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,688 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Select a safe numeric column
num_col <- names(df)[sapply(df, is.numeric)][1]
```

```
# Convert numeric column to factor
df$num_factor <- as.factor(df[[num_col]])
print(df)
```

```
# A tibble: 55,912 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
```

```
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Rename first level of factor safely
levels(df$num_factor)[1] <- "New"
print(df)
```

```
# A tibble: 55,912 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Reorder factor levels (alphabetical order)
df$num_factor <- factor(df$num_factor, levels = sort(levels(df$num_factor)))
print(df)
```

```
# A tibble: 55,912 × 31
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 23 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Cut numeric column into 3 categories (bins)
df$bins <- cut(df[[num_col]], breaks = 3)
print(df)
```

```
# A tibble: 55,912 × 32
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 24 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Create category using ifelse condition
mean_val <- mean(df[[num_col]], na.rm = TRUE)
df$flag2 <- ifelse(df[[num_col]] > mean_val, "Above Avg", "Below Avg")
print(df)
```

```
# A tibble: 55,912 × 33
```

	latitude	first_column	categories	city	country	longitude	name	postalCode
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 25 more variables: province <chr>, reviews.date <dtm>,
# reviews.dateAdded <dtm>, reviews.doRecommend <lgl>, reviews.id <lgl>,
# reviews.rating <dbl>, reviews.text <chr>, reviews.title <chr>,
# reviews.userCity <chr>, reviews.username <chr>, reviews.userProvince <chr>,
# id <chr>, dateAdded <dtm>, dateUpdated <dtm>, primaryCategories <chr>,
# keys <chr>, reviews.dateSeen <chr>, reviews.sourceURLs <chr>, ...
```

```
# Change all column names to uppercase
names(df) <- toupper(names(df))
print(df)
```

```
# A tibble: 55,912 × 33
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 25 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```

```
# Select a safe numeric column
```

```
num_col <- names(df)[sapply(df, is.numeric)][1]
```

```
# 1. Recodes values into High/Low categories
```

```
df$category <- ifelse(df[[num_col]] > mean(df[[num_col]]), na.rm = TRUE), "High", "Low")
print(df)
```

```
# A tibble: 55,912 × 34
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 26 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```



```
# 2. Converts numeric column into factor
df$num_factor <- as.factor(df[[num_col]])
print(df)
```

```
# A tibble: 55,912 × 35
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 27 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```

```
# 3. Creates bins (groups values into ranges)
df$bins <- cut(df[[num_col]], breaks = 3)
print(df)
```

```
# A tibble: 55,912 × 36
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 28 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```

```
# 4. Replaces negative values with 0
df[[num_col]][df[[num_col]] < 0] <- 0
```

```
print(df)
```

```
# A tibble: 55,912 × 36
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 28 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,  
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,  
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,  
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,  
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,  
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```

```
# 5. Rounds numeric values
```

```
df[[num_col]] <- round(df[[num_col]], 2)  
print(df)
```

```
# A tibble: 55,912 × 36
```

	LATITUDE	FIRST_COLUMN	CATEGORIES	CITY	COUNTRY	LONGITUDE	NAME	POSTALCODE
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
2	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
3	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
4	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
5	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
6	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
7	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
8	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
9	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126
10	45.4	Riviera San Nic...	Hotels	Mabl...	US	12.4	Hote...	30126

```
# i 55,902 more rows
```

```
# i 28 more variables: PROVINCE <chr>, REVIEWS.DATE <dtm>,  
# REVIEWS.DATEADDED <dtm>, REVIEWS.DORECOMMEND <lgl>, REVIEWS.ID <lgl>,  
# REVIEWS.RATING <dbl>, REVIEWS.TEXT <chr>, REVIEWS.TITLE <chr>,  
# REVIEWS.USERCITY <chr>, REVIEWS.USERNAME <chr>, REVIEWS.USERPROVINCE <chr>,  
# ID <chr>, DATEADDED <dtm>, DATEUPDATED <dtm>, PRIMARYCATEGORIES <chr>,  
# KEYS <chr>, REVIEWS.DATESEEN <chr>, REVIEWS.SOURCEURLS <chr>, ...
```